

Project Synopsis

Create an Amazon S3 Bucket and Upload a File

Team Details

S.No.	Name	Roll No.
1	Kapil Jadon	2415000767
2	Ankita Singh	2415000239
3	Raj Pratap Singh	2415001253
4	Ankush	2415800013

Branch: Computer Science & Engineering (CSE)

Stream / Specialization: Cloud & DevOps

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Aim

To create an Amazon S3 bucket, upload an HTML file, configure public access, and access the uploaded file using its Object URL.

Prerequisites

- AWS Account
- AWS Management Console
- Internet Connectivity
- A text editor (Notepad or Visual Studio Code)

Methodology / Procedure Overview

1. Open Amazon S3

Signed in to the AWS Management Console, searched for the S3 service, and navigated to it to begin creating a new bucket.

2. Create an S3 Bucket

A new bucket named lakhan-s3-bucket-`<unique-name>` was created in the Asia Pacific (Mumbai) – ap-south-1 region, with default object ownership settings and public access unblocked (with acknowledgment) to allow later public read configuration.

3. Create an HTML File

A simple HTML file (index.html) was created using Notepad, containing a heading, subheading, and a short paragraph describing the project.

4. Upload the File to S3

The index.html file was uploaded to the created bucket, and the upload was verified as successful.

5. Configure Public Read Access

A bucket policy was applied in JSON format to grant public GetObject permission on all objects in the bucket, enabling anyone with the link to view the file. The policy used was:

```
{  "Version": "2012-10-17",  "Statement": [    {      "Sid": "PublicReadAccess",      "Effect": "Allow",      "Principal": "*",      "Action": "s3:GetObject",      "Resource": "arn:aws:s3:::lakhan-s3-bucket-123/*"    }  ] }
```

6. Access the Uploaded File

The Object URL of the uploaded index.html file was copied and opened in a new browser tab, confirming that the HTML page loaded successfully and was publicly accessible.

Result

An Amazon S3 bucket was successfully created, an HTML file was uploaded, public read access was configured, and the uploaded file was successfully accessed using its Object URL.

Applications

- Hosting static websites (HTML, CSS, JS) directly from an S3 bucket.
- Storing and distributing images, videos, and documents at scale.
- Backup and disaster recovery storage for applications and databases.
- Serving as an origin for content delivery networks (e.g., Amazon CloudFront).
- Data lake storage for analytics and big data processing pipelines.
- Sharing files publicly or privately with fine-grained access control.

Advantages

- Highly scalable storage with virtually unlimited capacity.
- Pay-as-you-go pricing with no upfront infrastructure cost.
- High durability and availability backed by AWS's global infrastructure.
- Simple integration with other AWS services (CloudFront, Lambda, EC2, etc.).
- Fine-grained access control through bucket policies and IAM.
- Easy accessibility via Object URLs, AWS CLI, SDKs, or REST API.

Limitations

- Making a bucket or object public requires care, as misconfigured policies can expose sensitive data.
- Not suited for scenarios requiring low-latency block or file storage (e.g., databases).
- Object size is limited to 5 TB per upload, with additional configuration needed for very large files.
- Costs can grow with high request rates, data transfer, and storage class choices if not monitored.
- Bucket names must be globally unique across all AWS accounts, which can complicate naming.

Future Scope

- Integrating the S3 bucket with Amazon CloudFront for faster global content delivery.
- Enabling versioning and lifecycle policies for better data management and cost optimization.
- Automating deployment using AWS CLI, SDKs, or Infrastructure as Code tools (e.g., Terraform, CloudFormation).
- Adding a custom domain name and HTTPS support for the hosted website.
- Implementing logging and monitoring using AWS CloudTrail and CloudWatch.

Conclusion

This project demonstrated how to create an Amazon S3 bucket, upload a file, configure public access, and access the uploaded file through its Object URL. Amazon S3 provides a simple and reliable way to store and share files over the internet, making it a foundational service for cloud storage and static content hosting.